PLUG PULLER · TWO-SIDED PULLER

Full guide, two-sided puller

Everything about the two-sided puller: getting the file, measuring, every dial with a picture, printing, assembly, use and care, what to do when a print does not fit, every warning, and how the advanced dials work.

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Which tool this is

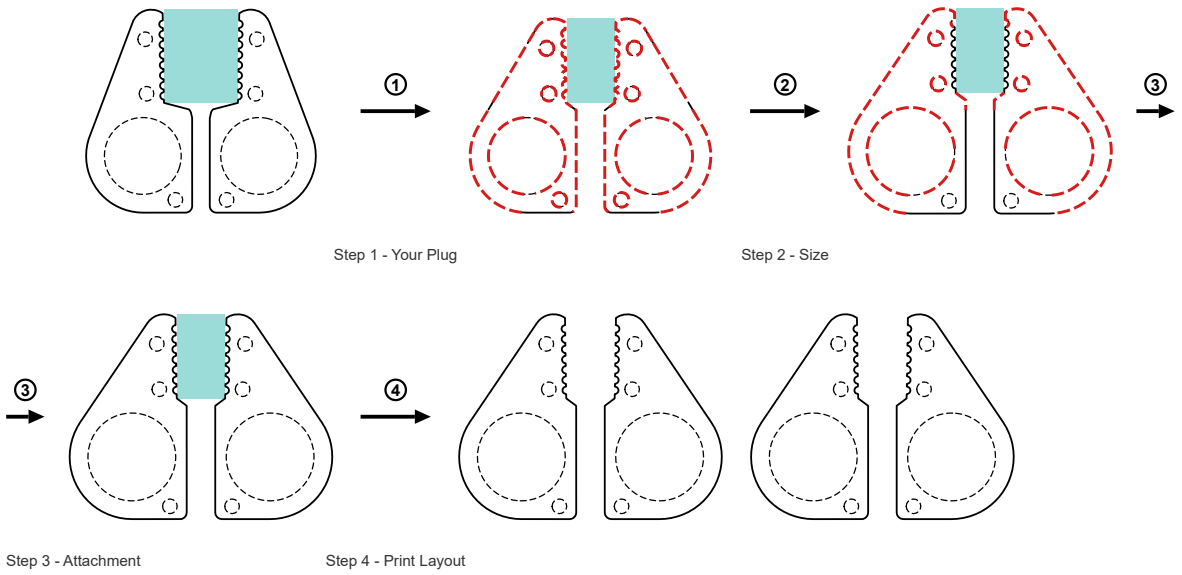
This guide is for the two-sided puller, src/Plug_Puller_Two_Sided.scad: two serrated plates that zip-tie around the plug and close across it, for a plug 24 mm thick or more and for a plug held from both sides, a USB-C tip or a round extension-cord plug. A thinner wall plug is the one-sided puller's job, and that tool has its own guide.

The Customizer's four steps are your plug, your size, the attachment and the print layout. This full guide covers getting the file, measuring your plug and your hand or matching a card, every dial with a picture, printing, assembly, use and care, what to do when a print does not fit, every warning, and how the advanced dials work.

> If red text appears beside the part in the preview, read it: it names the measurement to fix, and the part will not fit until it is gone.

How to read the pictures

- **Black:** the tool
- **Red dashed:** the edges this dial moved
- **Teal:** your plug
- ① the numbers match the key beside the picture



Five stages of the two-sided puller, left to right, the top row first, the plug end at the top; arrows numbered 1 to 4 show the steps, and red dashes mark the edges each step moved. Start, the defaults: the tool as the file opens, with a 20 mm wide plug in teal. Step 1 - Your Plug: plug length 23 mm, plug width at both ends 13 mm, cord thickness 7 mm, plug sides Rounded sides; red on the arms, the finger lobes and the zip stations. Step 2 - Size: hand size Large; red on the arms, the finger lobes and the zip stations. Step 3 - Attachment: attachment Zip ties; no strap slot on a plug this short, so nothing moved. Step 4 - Print Layout: both plates side by side in one file, what you print; nothing marked.

- Step 1: type your plug's numbers and pick Rounded sides or Flat sides
- Step 2: pick your hand size
- Step 3: pick how it attaches; a plug this short gets no strap slot
- Step 4: both plates in one file

Get the file

The two-sided puller is the file `src/Plug_Puller_Two_Sided.scad`. There are two ways to open it and fill in its form: in your web browser with nothing to install, or in the free OpenSCAD program on your computer. Both show the same form, called the Customizer.

In your browser

The OpenSCAD Assistive Forge is a version of OpenSCAD that runs in your browser, on a computer or a phone, built for keyboard and screen reader use. Nothing is uploaded: your numbers stay on your device.

1. Open the two-sided puller in the Assistive Forge. On a first visit Forge asks which interface you want: choose **Assistive Forge**, then **Download & Continue**, and it downloads its engine once.
2. Forge offers to keep a copy of the project in your browser. Choose **Save My Copy**: the tool is then listed on Forge's main page whenever you come back, no link needed.
3. The form shows the four steps' dials first. The rest sit behind one button, **Show all parameters**.

If the link cannot load, Forge says why and offers **Try again**. You can also download the whole tool as one file, `dist/Plug_Puller_Two_Sided_SingleFile.scad`, and open it from Forge's main page, under Open or start a project.

On your computer

OpenSCAD is the free program that turns your numbers into a printable file. You use one panel of it and never touch the code.

1. Download OpenSCAD from https://openscad.org/downloads.html: the **Development Snapshot** for your system. This project is tested with the snapshot of 2026-01-03; any recent snapshot works. The regular release works too, only slower.
2. Get the project: on its GitHub page press the green **Code** button, then **Download ZIP**, and unzip it anywhere. Or download only `dist/Plug_Puller_Two_Sided_SingleFile.scad`, the whole tool in one file.
3. Open `src/Plug_Puller_Two_Sided.scad` (or the single file) in OpenSCAD: double-click it, or use **File ▸ Open**. A wall of code appears in an editor pane. Ignore it; you will not touch it.
4. Show the Customizer, the form you type into: in the **View** menu, make sure **Hide Customizer** is unchecked. In older versions it is **Window ▸ Customizer**.

The form's sections read top to bottom in the order you decide things: **Step 1 - Your Plug**, **Step 2 - Size**, **Step 3 - Attachment**, then **Step 4 - Print Layout**. Everything below Step 4 is optional.

Working in the browser

Forge and the desktop program show the same form and make the same tool. What differs:

- Forge draws the tool as it will print, with any red warning text beside it. It does not draw the see-through plug, or the green and orange notes of the desktop program's preview.
- A render takes longer in the browser than on the desktop, and longest on a phone. Forge says **Preview ready** under the picture when it is done.
- On a phone the screen is tight. The parameter search finds any dial by its name, and searching also brings back the dials behind **Show all parameters**.
- Forge can be installed as an app that works offline: use your browser's install button at the right end of the address bar.
- To send your numbers to someone, or to keep them, use Forge's **Copy Link**: the link carries the values you changed, and opening it puts them back.

Problem	What to do
"The shared project could not be opened"	Forge names the reason. Press **Try again**; if it keeps failing, download the single file and open it from Forge's main page.
A dial you want is not on the form	Press **Show all parameters**. The four steps' dials are shown first; the rest sit behind that button.
Red text beside the tool in the preview	A measurement cannot work, and the text names it. Fix it before you export: the red text is part of the model.

Measure your plug and your hand

Everything here is in mm: a US plug is about 25 mm wide, so a 1 on your paper means you measured in inches. You need a caliper or a ruler with mm marks, the plug in its outlet, and your own hand only if you pick Measure my hand. Print the measuring form at 100 % and fill it in as you go: the sections below are the form's rows, in the same order, and the card names (R1, C1, F1 / F2) are the cards of the printed measuring stencil, described under Match a card instead of measuring.

The two-sided puller's widths are the size the two plates close across, so measure across the plug the way the plates will grip it. The thickness pair, the wall plate style and the hand width belong to the one-sided puller only.

Form row	What to measure	How	Typical	Stencil card
2	**Plug length** measure_plug_length	With the plug in the outlet, ruler from the wall plate face to the plug's back face. On a laptop or charger plug, measure from the device's edge.	15 to 55 mm; example 32.5 mm	card R1
3	**Plug width at the prong end** measure_plug_width_prong_end	Caliper across the plug body just behind the prongs, the size the two plates close across. On a USB-C or charger tip, just behind the metal tip; on a round plug it is the diameter.	10 to 40 mm; example 20 mm	card R1
4	**Plug head length** measure_plug_head_length	Ruler along the plug from the prong end to where the plug starts to narrow. If it narrows right from the prong end, type 0.	0 to 25 mm; example 15.2 mm	card R1
5	**Plug width at the cord end** measure_plug_width_cord_end	The same direction across the plates, measured at the plug's back end where the cord comes out. A rubber strain relief counts as part of the plug.	10 to 40 mm; example 11.1 mm	card R1
6	**Cord thickness** measure_cord_thickness	Caliper across the cord just behind the plug, the way the widths run: a flat cord's wide side, a round cord's diameter. On the stencil, it is the smallest C1 slot that takes the cord's wide side.	3 to 10 mm; example 7 mm	card C1
7	**Plug sides** plug_sides	Look at the plug body where the plates will grip it. Rounded sides for a round cord plug, a USB-C or charger tip; Flat sides for a boxy plug.	Choices: Rounded sides, Flat sides	none
10	**Finger width** measure_finger_width	Caliper across the widest knuckle of your middle finger, the finger that goes into the pull hole. With the stencil, find the smallest F1 or F2 hole your finger passes through comfortably and subtract 5 from its number. No caliper? Take a ring that fits that finger snugly, measure the ring's inner diameter in mm and add 1.5 mm.	16 to 24 mm; example 22 mm	card F1 / F2
12	**Strap width** strap_width	Read the width on the strap's packaging. ONE-WRAP comes in 10, 13, 16, 20 and 25 mm.	10 to 25 mm; example 15 mm	none

In every picture: black = the tool; teal = your plug; red dashed = the edges this dial moved; the numbers match the key beside the picture.

> Sanity check: each plug width is a two-digit number, roughly 12 to 45, and the prong-end width is usually the bigger one; the finger width is roughly 14 to 32. A number like 1.3 is inches: measure again with the mm side.

When you measure for someone else, a relative or a client, measure their hand for the finger and hand rows and their outlet and plug for the plug rows. Doubtful between two values? Round up: a slightly roomy fit works, a tight one does not.

The measuring form

> The form is printed at true size on page 00, at the end of this guide; the paper stencil sheets follow it. Print the form at 100 % (actual size, never fit to page) and check its 50 mm bar with a ruler before you trust it. Fill in the blanks top to bottom as you measure, then type the numbers into the Customizer in the same order.

Match a card instead of measuring

The measuring stencil is a set of thin printed cards. Each has a two-letter name raised on its face, so you can find it by touch. Hold your plug in the plug cards: if it fills one card's openings, snug with no big gaps, that card names the preset to pick in Step 1, and you type no numbers at all.

Card	What it is	What it answers
P1	The flat 2-prong lamp plug (NEMA 1-15)	the preset `Flat 2-prong lamp plug - NEMA 1-15`
P2	The standard 3-prong plug (NEMA 5-15)	the preset `Standard 3-prong plug - NEMA 5-15`
P3	The heavy-duty extension cord plug (NEMA 5-15)	the two-sided puller's preset `Heavy-duty extension cord - NEMA 5-15`
P4	The wide 2-prong appliance plug (NEMA 1-15)	the one-sided puller's preset `Wide 2-prong appliance plug - NEMA 1-15`
R1	A ruler: raised mm ticks, numbers every 10 mm, and a notch every 10 mm you can count by touch	the plug's length, widths and thicknesses
C1	A cord gauge: open slots 3 to 9 mm wide along one edge, slid onto the cord from the side	the cord thickness: the smallest slot that slips over the cord
F1, F2	Finger holes 15 to 25 mm across (F1) and 26 to 32 mm (F2)	your finger width: the smallest hole your middle finger passes through comfortably, minus 5

Each P card has three openings: W for the plug's width, T for its thickness, and an open slot that slides onto the cord. If no card fits, your plug is between presets: measure it instead. The measured tool fits better than any preset.

Printing the cards. Print `stl/Measuring-Stencil/Visual/Measuring-Stencil_Visual_All-Cards.stl`: 1.2 mm thick cards, no supports, any rigid filament, packed onto sheets for a 200 by 200 mm bed. The tactile version, `stl/Measuring-Stencil/Tactile/`, prints every label as a raised character at ADA size and adds a Grade 2 braille title flap to every card. Or print one card at a time from `stl/Measuring-Stencil/`. For a smaller bed, open `Measuring_Stencil.scad` in OpenSCAD, set `bed_width` and `bed_depth` to your bed, and export `part_index` 1, then 2, and so on, one sheet at a time. `part_index` 0 previews every sheet at once; do not print that one.

Folding the tactile flaps. The tactile version is also `label_mode` set to Tactile in the file. Each flap prints leaning back, held by thin fins. Snap the fins off, then fold the flap away from the card until it lies flat, so the braille lands face

up beyond the card's edge. Fold once, gently; a PETG or PP hinge folds more reliably than PLA.

No 3D printer yet? The last pages of the printed quick start and full guide are paper versions of the cards at true size: the plug outlines, a 100 mm ruler and the finger circles. Print them at 100 % (actual size, never fit to page) and check the 50 by 50 mm square with a ruler before you trust them. On their own they are `stencil-sheet.svg` and `stencil-sheet-2.svg`.

Try it on paper first

Every preset at every size has a printable sheet with one plate's exact outline at true size, its dimensions in mm, a 50 by 50 mm square to check the print scale, and the Customizer settings that make it. Print one at 100 % (actual size, never fit to page) and measure the square with a ruler: it must be exactly 50 by 50 mm, or the print was scaled. Cut out the outline along the solid line and poke through the two finger circles. Hold it against your plug: the plug body sits between the two serrated edges with the cord in the channel at the bottom, and the real tool is two of these plates. Try the finger holes; if they feel wrong, try the next size's sheet.

Plug preset	Small	Medium	Large
Heavy-duty extension cord (NEMA 5-15)	sheet	sheet	sheet
USB-C laptop tip	sheet	sheet	sheet
Flat 2-prong lamp plug (NEMA 1-15)	sheet	sheet	sheet
Standard 3-prong plug (NEMA 5-15)	sheet	sheet	sheet

Every sheet, for both tools, is also in one printable PDF with an index: `docs/Plug_Puller_Outline_Sheets.pdf`. The sheets cover the presets only; a plug or a hand between sizes is better served by measuring.

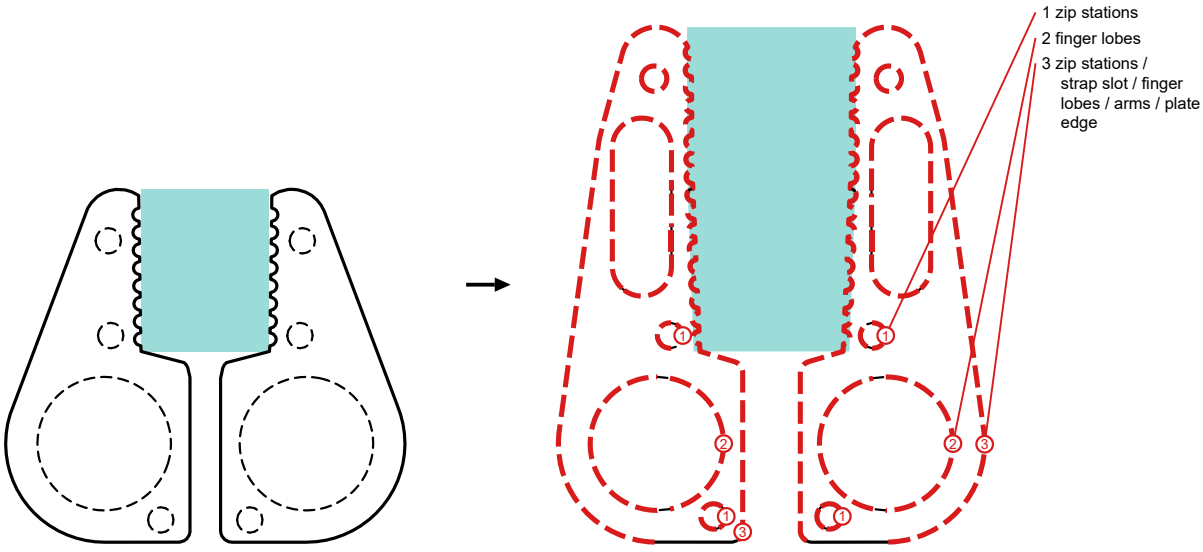
Step 1 - Your Plug

Pick a plug preset, or leave it on Measure my plug and type your plug's numbers; then pick Rounded sides or Flat sides.

Plug preset

Customizer name: **plug_preset**

Fills in the plug's length, both widths, the head length, the cord and the sides from that plug's measurements, so the arms, the grip gap, the cord channel and the plate length all take that plug's shape at once.



Measure my plug
→ Heavy-duty extension cord - NEMA 5-15

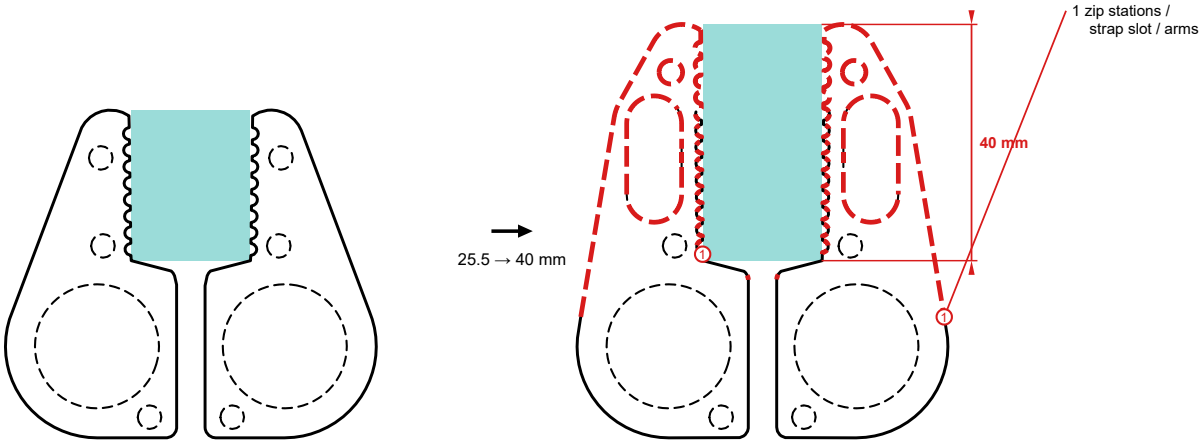
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: plug preset set to Heavy-duty extension cord - NEMA 5-15. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the finger lobes: the two rounded lobes in the lower half. 3, the zip stations, the strap slot, the finger lobes, the arms and the plate edge.

Default	Measure my plug
Choices	Measure my plug, Heavy-duty extension cord - NEMA 5-15, USB-C laptop tip, Flat 2-prong lamp plug - NEMA 1-15, Standard 3-prong plug - NEMA 5-15
Moves	arms, finger lobes, plate edge, strap slot, zip stations

Plug length

Customizer name: **measure_plug_length**

Lengthens both arms so the teeth cover the whole plug body, and moves the zip-tie stations and the strap slot with them.



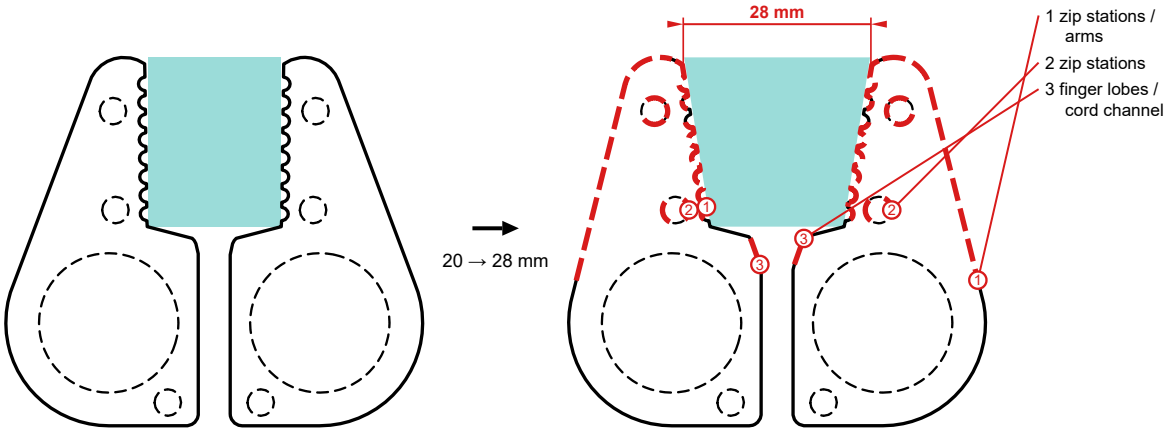
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: plug length at 40 mm. The red line marks the plug's length and reads 40 mm, the number you type. Marked in red: 1, the zip stations, the strap slot and the arms: the three small holes along each arm; the long slot in each arm; the two toothed arms in the upper half.

Default	25.5 mm
Range	12 to 85 mm, in steps of 0.5 mm
Moves	arms, strap slot, zip stations

Plug width at the prong end

Customizer name: **measure_plug_width_prong_end**

Opens or closes the gap between the arms at their tips, where the plug's prong end sits.



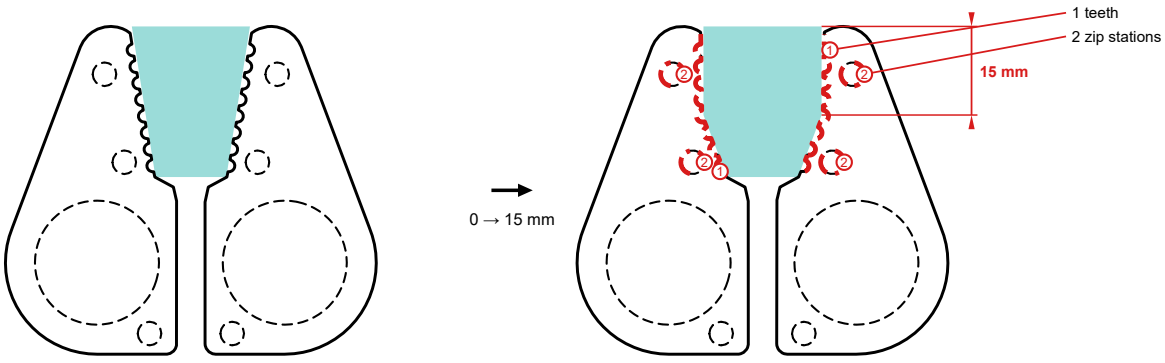
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: plug width at the prong end at 28 mm. The red line marks the gap between the arm tips and reads 28 mm, the number you type. Marked in red: 1, the zip stations and the arms. 2, the zip stations: the three small holes along each arm. 3, the finger lobes and the cord channel.

Default	20 mm
Range	5 to 40 mm, in steps of 0.5 mm
Moves	arms, cord channel, finger lobes, zip stations

Plug head length

Customizer name: **measure_plug_head_length**

Keeps the gap between the arms at its prong-end width for the length of the plug's head, so the gap narrows only behind the head.



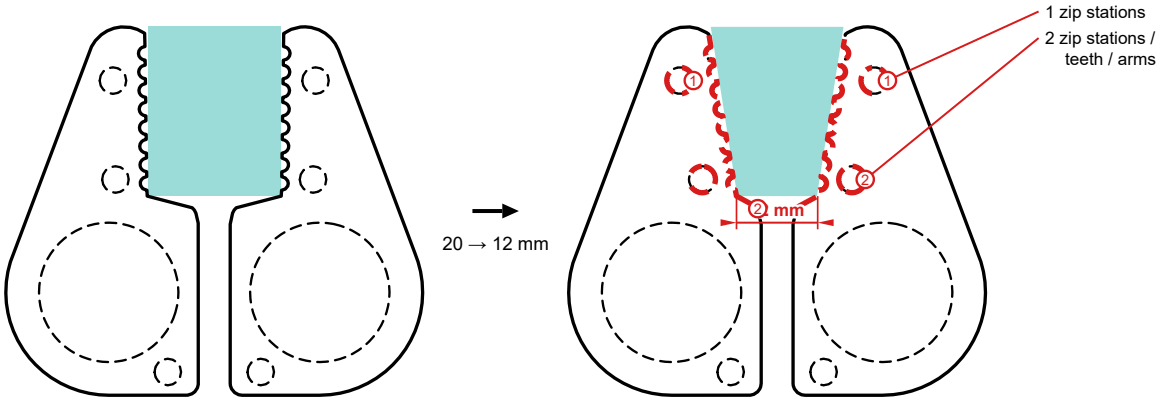
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug width at the cord end set to 12, a 20 mm wide plug in teal. Right: plug head length at 15 mm. The red line marks the plug's head length and reads 15 mm, the number you type. Marked in red: 1, the teeth: the serrated inner edges of the arms. 2, the zip stations: the three small holes along each arm.

Default	0 mm
Range	0 to 60 mm, in steps of 0.5 mm
Moves	teeth, zip stations

Plug width at the cord end

Customizer name: **measure_plug_width_cord_end**

Opens or closes the gap between the arms at the plug's back end, so the arms taper to match the plug.



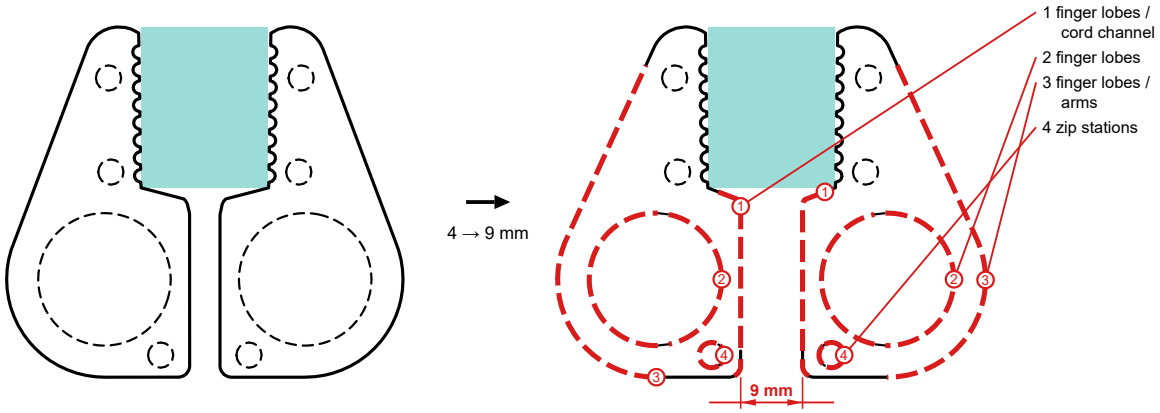
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: plug width at the cord end at 12 mm. The red line marks the gap between the arms at the cord end and reads 12 mm, the number you type. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the zip stations, the teeth and the arms: the three small holes along each arm; the serrated inner edges of the arms; the two toothed arms in the upper half.

Default	20 mm
Range	5 to 40 mm, in steps of 0.5 mm
Moves	arms, teeth, zip stations

Cord thickness

Customizer name: **measure_cord_thickness**

Widens the cord channel between the finger holes, and the finger lobes move outward with it.



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: cord thickness at 9 mm. The red line marks the cord channel and reads 9 mm, the number you type; the channel is drawn 9.8 mm wide, so the cord slips in. Marked in red: 1, the finger lobes and the cord channel. 2, the finger lobes: the two rounded lobes in the lower half. 3, the finger lobes and the arms. 4, the zip stations: the three small holes along each arm.

Default	4 mm
Range	1.5 to 12 mm, in steps of 0.5 mm
Moves	arms, cord channel, finger lobes, zip stations

Plug sides

Customizer name: **plug_sides**

Removes the sloped cradle from the arms' gripping edges so the teeth bite straight along a flat-sided plug.



Rounded sides
→ Flat sides
section at y = 45 mm

Two vertical slices of the two-sided puller at y = 45 mm, before left and after right, the top face up. Left: the defaults, the plug in teal in the cut. Right: plug sides set to Flat sides. Marked in red: 1, the teeth and the arms: the serrated inner edges of the arms; the two toothed arms in the upper half. 2, the arms: the two toothed arms in the upper half.

Default	Rounded sides
Choices	Rounded sides, Flat sides
Moves	arms, teeth

Show the see-through plug

Customizer name: **show_plug_preview**

Shows or hides the see-through plug in the preview and changes nothing in the printed plates.

> **No picture.** Changes no shape: the see-through plug is a preview aid and is never exported.

Default	On
Choices	On or off (a check box)

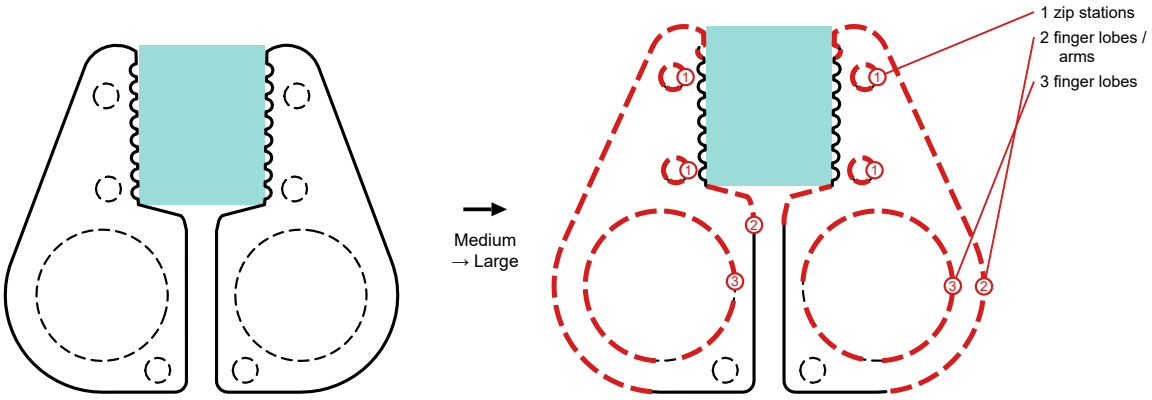
Step 2 - Size

Pick your hand size, or pick Measure my hand and type your finger width.

Hand size

Customizer name: **size**

Widens both finger holes and their lobes, so the whole plate grows around them.



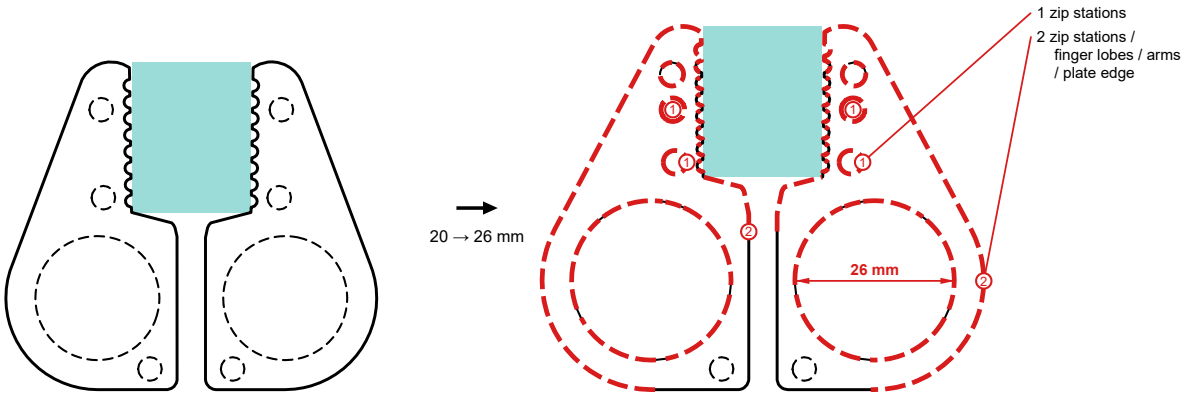
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: hand size set to Large. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the finger lobes and the arms: the two rounded lobes in the lower half; the two toothed arms in the upper half. 3, the finger lobes: the two rounded lobes in the lower half.

Default	Medium
Choices	Small, Medium, Large, Measure my hand
Moves	arms, finger lobes, zip stations

Finger width

Customizer name: **measure_finger_width**

Widens both finger holes and their lobes.



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Measure my hand, a 20 mm wide plug in teal. Right: finger width at 26 mm. The red line marks a finger hole and reads 26 mm, the number you type; the hole is drawn 27 mm across, wider than the finger. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the zip stations, the finger lobes, the arms and the plate edge.

Default	20 mm
Range	14 to 32 mm, in steps of 0.5 mm
Moves	arms, finger lobes, plate edge, zip stations

> **Note:** Only acts when size is Measure my hand.

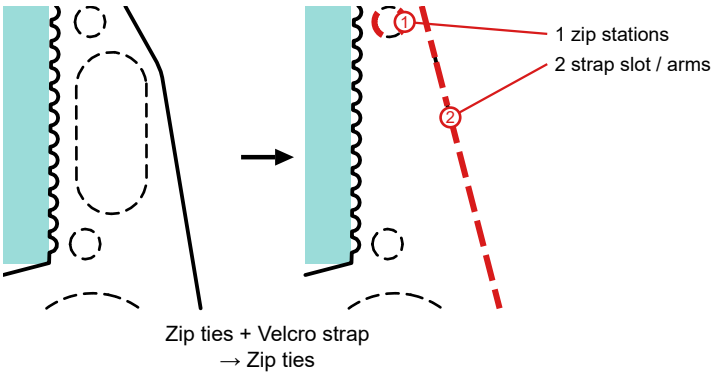
Step 3 - Attachment

Pick how the tool attaches.

Attachment

Customizer name: **attachment**

Removes the strap slot from each arm, while the zip-tie stations stay because they hold the plates together.



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40, a 20 mm wide plug in teal. Right: attachment set to Zip ties. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the strap slot and the arms: the long slot in each arm; the two toothed arms in the upper half.

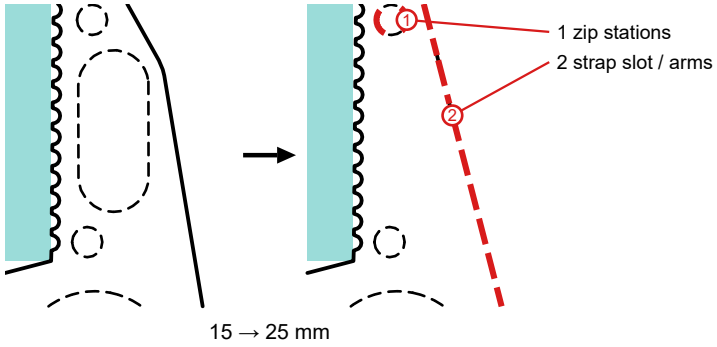
Default	Zip ties + Velcro strap
Choices	Zip ties + Velcro strap, Zip ties
Moves	arms, strap slot, zip stations

> **Note:** Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot.

Strap width

Customizer name: **strap_width**

Sets the strap the arm slot must clear, and when the slot's window between the zip-tie stations is shorter than the strap plus 1.5 mm the slot is left out.



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40, a 20 mm wide plug in teal. Right: strap width at 25 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the strap slot and the arms: the long slot in each arm; the two toothed arms in the upper half.

Default	15 mm
Range	10 to 25 mm, in steps of 1 mm
Moves	arms, strap slot, zip stations
Red warnings	STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP

Note: Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot. At 25 mm the slot leaves too, because the plug is too short for one, so the after picture has no slot and no dimension line.

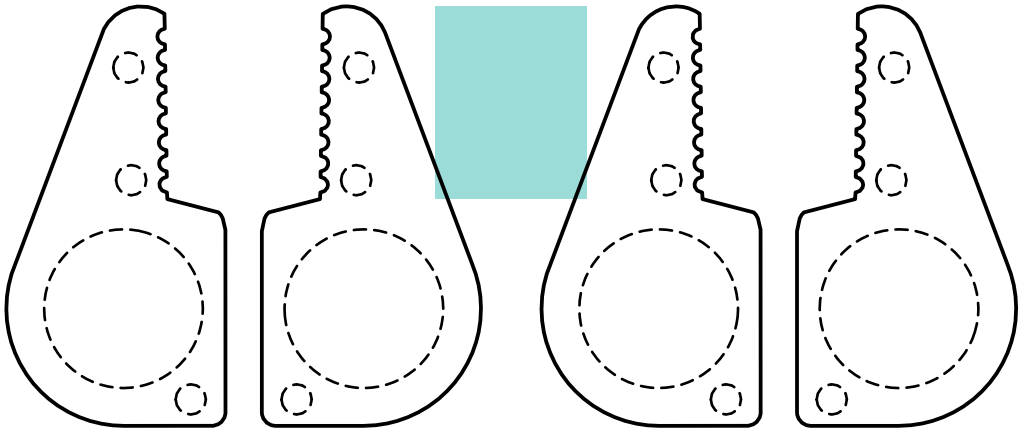
Step 4 - Print Layout

Pick whether the file holds both plates or one.

Print layout

Customizer name: **print_layout**

Puts both identical plates side by side in one file, or just one plate.



Both plates
→ One plate

One top view of the two-sided puller with both plates side by side, the plug end at the top: what the file prints at Both plates. At One plate it prints one plate. Nothing is marked in red: this dial changes the layout, not the plate.

Default	Both plates
Choices	Both plates, One plate
Moves	arms, finger lobes, plate edge, zip stations

The optional dials

Everything below Step 4 in the Customizer is optional. These dials apply in every hand size, except the sections marked (Custom size only), which do nothing until Step 2's hand size is Custom. The sections follow in the Customizer's order; how they work together is explained after the troubleshooting sections.

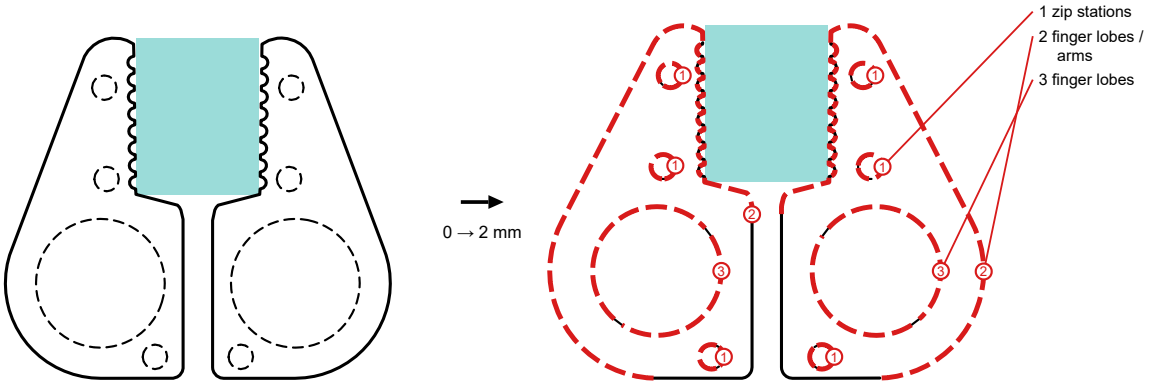
- **Advanced - Two-Sided Puller**, 26 dials: Extra wall on the plate, Plate thickness, Grip bite, Cradle depth, Grip clearance, Cord channel clearance, Finger hole fit, Finger lobe wall, Wall beside the cord channel, Tooth size, Tooth spacing, Tooth depth, Toothed zone start, Toothed zone length, Tip flare, Arm tip width, Outer edge rounding, Cable strip thickness, Zip-tie hole diameter, Zip-tie station placement, Zip-tie station 1 position, Zip-tie station 2 position, Zip-tie station 3 position, Wall between the teeth and the slot, Strap slot width, Strap slot length.
- **Advanced - Render Quality**, 1 dial: Render quality.

Advanced - Two-Sided Puller

Extra wall on the plate

Customizer name: **plate_wall_boost**

Thickens every wall around the finger holes, the cord channel, the zip-tie holes and the strap slot at once, so the plate outline grows and the openings shift to keep their walls.



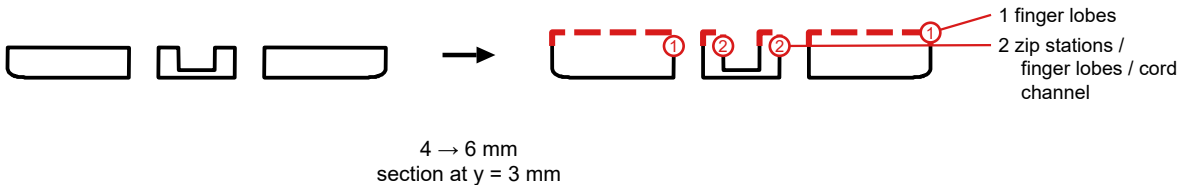
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: extra wall on the plate at 2 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the finger lobes and the arms: the two rounded lobes in the lower half; the two toothed arms in the upper half. 3, the finger lobes: the two rounded lobes in the lower half.

Default	0 mm
Range	0 to 5 mm, in steps of 0.25 mm
Moves	arms, finger lobes, zip stations

Plate thickness

Customizer name: **plate_thickness**

Thickens each plate, so the finished pair is twice as thick.



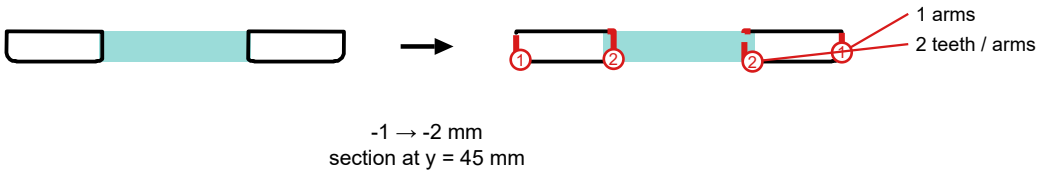
Two vertical slices of the two-sided puller at y = 3 mm, before left and after right, the top face up. Left: the defaults, the plug in teal in the cut. Right: plate thickness at 6 mm. Marked in red: 1, the finger lobes: the two rounded lobes in the lower half. 2, the zip stations, the finger lobes and the cord channel: the three small holes along each arm; the two rounded lobes in the lower half; the gap between the lobes at the bottom.

Default	4 mm
Range	2 to 8 mm, in steps of 0.25 mm
Moves	cord channel, finger lobes, zip stations
Red warnings	PLATE THINNER THAN 2MM - TOO FLIMSY

Grip bite

Customizer name: **plate_grip_bite**

Squeezes the plug harder or softer, because the arms close in by this much per side against the plug's width.



Two vertical slices of the two-sided puller at y = 45 mm, before left and after right, the top face up. Left: the defaults with plug sides set to Flat sides, the plug in teal in the cut. Right: grip bite at -2 mm. Marked in red: 1, the arms: the two toothed arms in the upper half. 2, the teeth and the arms: the serrated inner edges of the arms; the two toothed arms in the upper half.

Default	-1 mm
Range	-2 to 2 mm, in steps of 0.1 mm
Moves	arms, teeth
Red warnings	NO GRIP BITE - PLUG WONT BE HELD

> **Note:** Only acts when plug_sides is Flat sides or a plug preset is chosen.

Cradle depth

Customizer name: **plate_cradle_depth**

Slopes each arm's gripping edge from the mating face down to the outer face, so two plates form a cradle that centers a round plug.



2.5 → 0 mm
section at y = 45 mm

Two vertical slices of the two-sided puller at y = 45 mm, before left and after right, the top face up. Left: the defaults with plug sides set to Rounded sides, the plug in teal in the cut. Right: cradle depth at 0 mm. Marked in red: 1, the arms: the two toothed arms in the upper half. 2, the teeth and the arms: the serrated inner edges of the arms; the two toothed arms in the upper half.

Default	2.5 mm
Range	0 to 3.5 mm, in steps of 0.25 mm
Moves	arms, teeth
Red warnings	CRADLE SHALLOWER THAN ASKED - PLUG NARROW

> **Note:** Only acts when plug_sides is Rounded sides.

Grip clearance

Customizer name: **plate_grip_clearance**

Opens the gap between the arms where the plates meet, so a hard round plug can drop in flat before the cradle holds it.



0.5 → 2 mm
section at y = 45 mm

Two vertical slices of the two-sided puller at y = 45 mm, before left and after right, the top face up. Left: the defaults with plug sides set to Rounded sides, the plug in teal in the cut. Right: grip clearance at 2 mm. Marked in red: 1, the teeth and the arms: the serrated inner edges of the arms; the two toothed arms in the upper half. 2, the arms: the two toothed arms in the upper half.

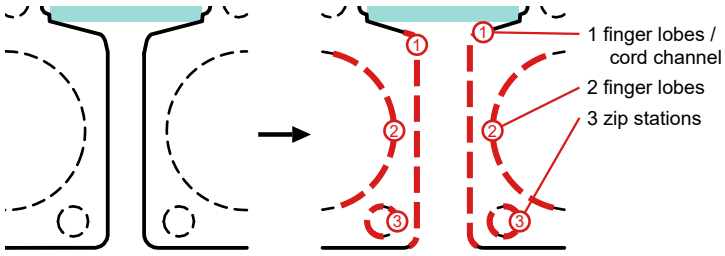
Default	0.5 mm
Range	0 to 2 mm, in steps of 0.1 mm
Moves	arms, teeth

> **Note:** Only acts when plug_sides is Rounded sides.

Cord channel clearance

Customizer name: **plate_cable_clearance**

Widens the cord channel between the finger holes beyond the cord's thickness.



0.8 → 3 mm

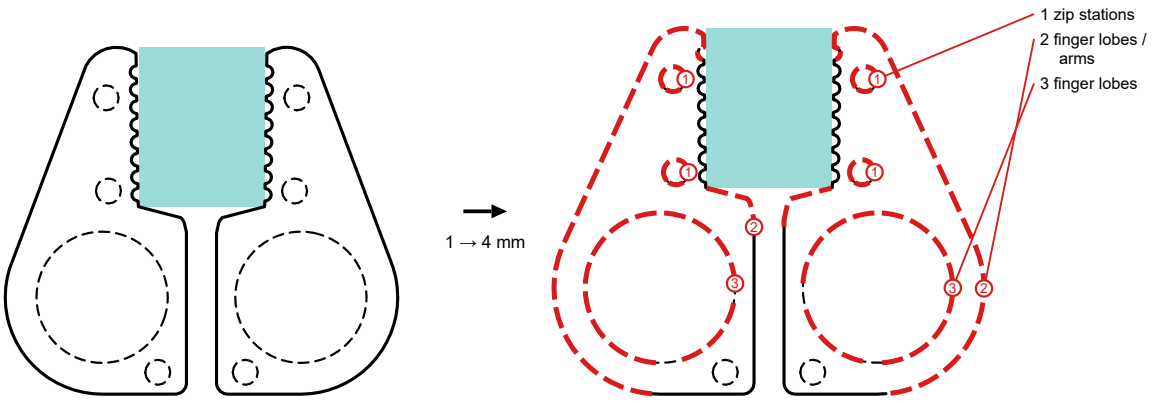
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: cord channel clearance at 3 mm. Marked in red: 1, the finger lobes and the cord channel: the two rounded lobes in the lower half; the gap between the lobes at the bottom. 2, the finger lobes: the two rounded lobes in the lower half. 3, the zip stations: the three small holes along each arm.

Default	0.8 mm
Range	0 to 5 mm, in steps of 0.1 mm
Moves	arms, cord channel, finger lobes, zip stations
Red warnings	CORD TOO THICK FOR CABLE CHANNEL PLUG NARROWER THAN THE CORD CHANNEL - ARMS CANNOT TOUCH IT

Finger hole fit

Customizer name: **plate_finger_fit**

Widens both finger holes beyond your finger width.



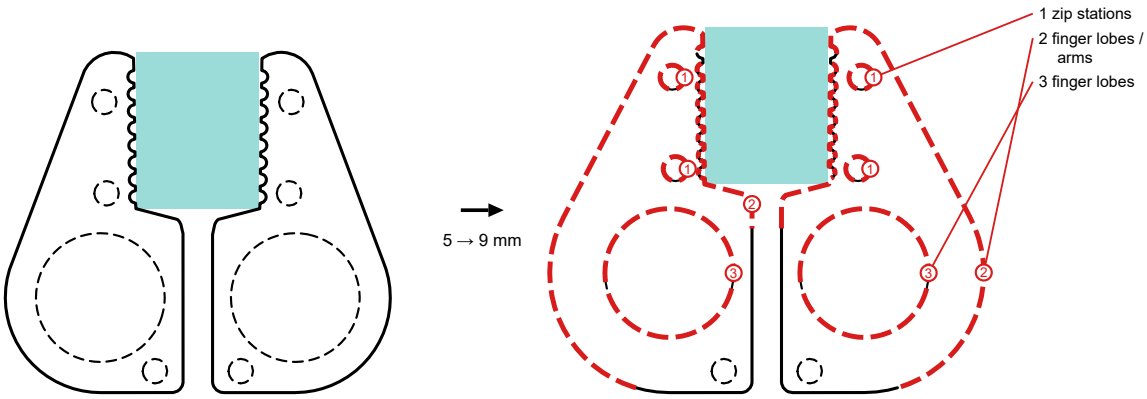
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: finger hole fit at 4 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the finger lobes and the arms: the two rounded lobes in the lower half; the two toothed arms in the upper half. 3, the finger lobes: the two rounded lobes in the lower half.

Default	1 mm
Range	0 to 8 mm, in steps of 0.25 mm
Moves	arms, finger lobes, zip stations

Finger lobe wall

Customizer name: **plate_finger_wall**

Thickens the wall around each finger hole, so the rounded lobes grow.



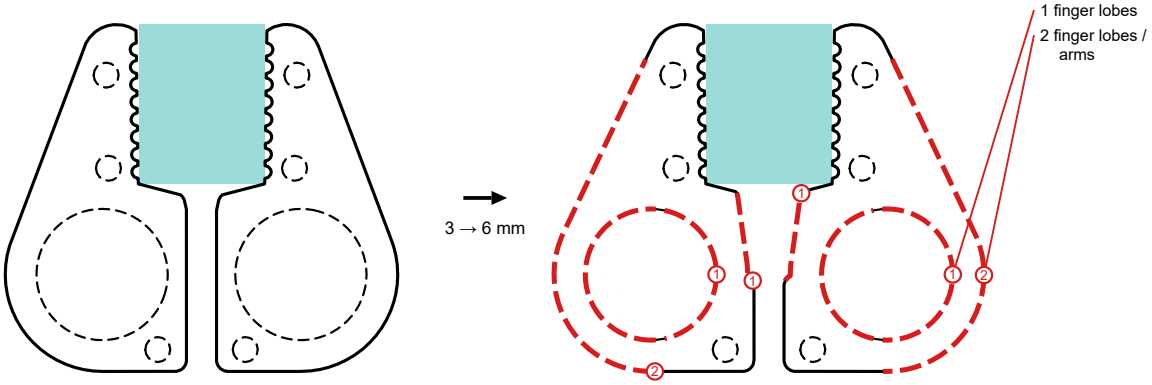
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: finger lobe wall at 9 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the finger lobes and the arms: the two rounded lobes in the lower half; the two toothed arms in the upper half. 3, the finger lobes: the two rounded lobes in the lower half.

Default	5 mm
Range	3 to 12 mm, in steps of 0.25 mm
Moves	arms, finger lobes, zip stations

Wall beside the cord channel

Customizer name: **plate_finger_inner_wall**

Thickens the wall between the cord channel and each finger hole, pushing the finger holes outward.



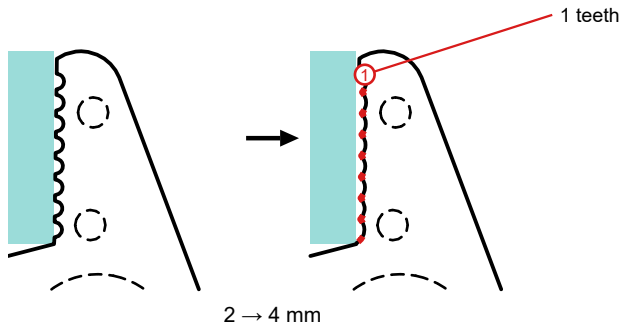
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: wall beside the cord channel at 6 mm. Marked in red: 1, the finger lobes: the two rounded lobes in the lower half. 2, the finger lobes and the arms: the two rounded lobes in the lower half; the two toothed arms in the upper half.

Default	3 mm
Range	1 to 8 mm, in steps of 0.25 mm
Moves	arms, finger lobes, zip stations

Tooth size

Customizer name: **plate_tooth_diameter**

Cuts bigger or smaller scallops into the arms' gripping edges.



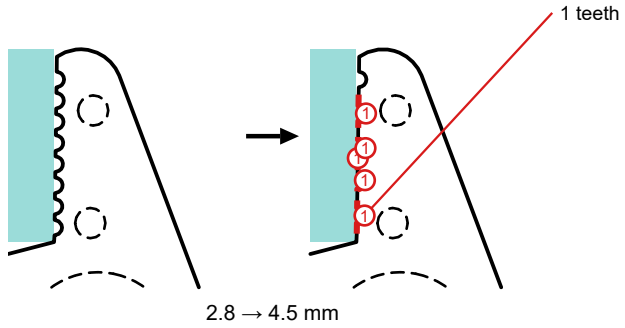
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: tooth size at 4 mm. Marked in red: 1, the teeth: the serrated inner edges of the arms. The plate edge, the arms, the finger lobes and the cord channel stay where they were.

Default	2 mm
Range	0 to 4 mm, in steps of 0.1 mm
Moves	teeth

Tooth spacing

Customizer name: **plate_tooth_pitch**

Spaces the teeth farther apart or closer together along the gripping edges.



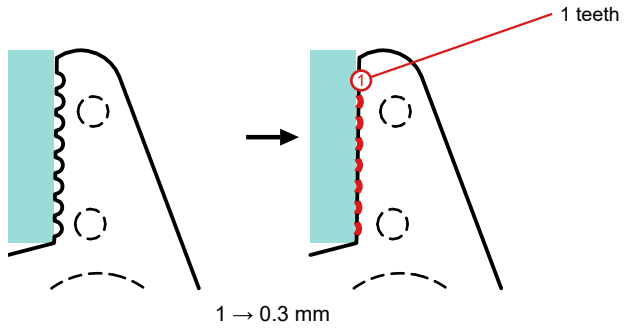
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: tooth spacing at 4.5 mm. Marked in red: 1, the teeth: the serrated inner edges of the arms. The plate edge, the arms, the finger lobes and the cord channel stay where they were.

Default	2.8 mm
Range	0.5 to 5 mm, in steps of 0.1 mm
Moves	teeth

Tooth depth

Customizer name: **plate_tooth_depth**

Cuts each tooth deeper or shallower into the gripping edge.



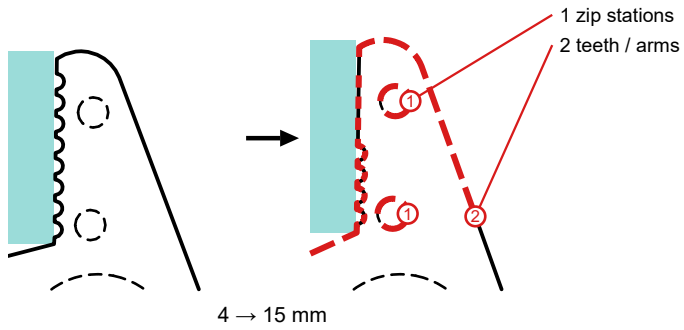
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: tooth depth at 0.3 mm. Marked in red: 1, the teeth: the serrated inner edges of the arms. The plate edge, the arms, the finger lobes and the cord channel stay where they were.

Default	1 mm
Range	0 to 1.5 mm, in steps of 0.05 mm
Moves	teeth

Toothed zone start

Customizer name: **plate_grip_zone_start**

Moves where the teeth begin, measured back from the arm tips, and the arms lengthen when the zone needs the room.



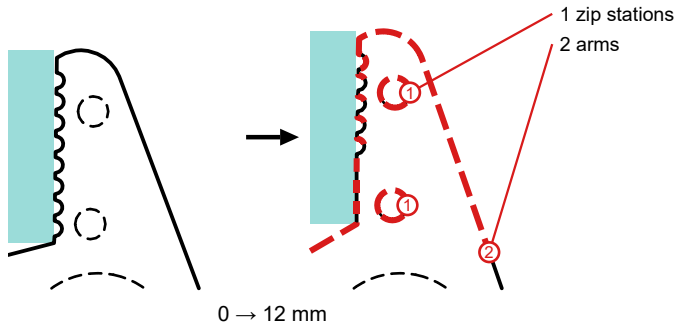
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: toothed zone start at 15 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the teeth and the arms: the serrated inner edges of the arms; the two toothed arms in the upper half.

Default	4 mm
Range	0 to 25 mm, in steps of 0.5 mm
Moves	arms, teeth, zip stations

Toothed zone length

Customizer name: **plate_grip_zone_length**

Sets how far the teeth run along each arm instead of covering the whole plug body.



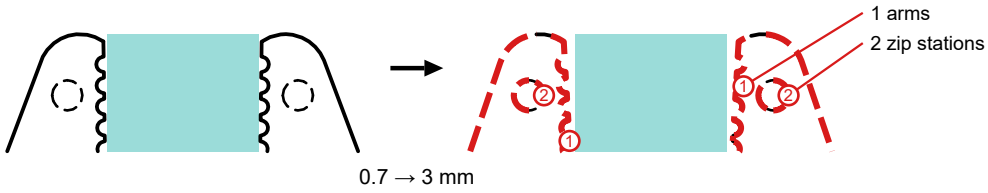
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: toothed zone length at 12 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the arms: the two toothed arms in the upper half.

Default	0 mm
Range	0 to 60 mm, in steps of 1 mm
Moves	arms, zip stations

Tip flare

Customizer name: **plate_tip_flare**

Opens the gap wider right at the arm tips so the plug head can enter before the teeth bite.



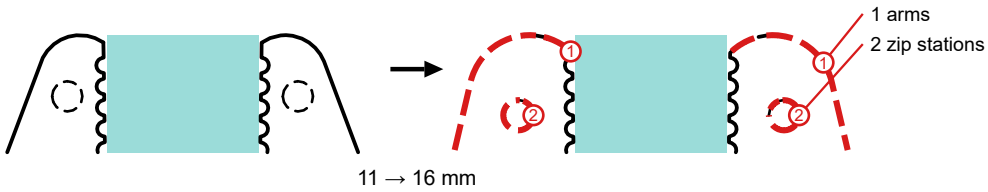
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: tip flare at 3 mm. Marked in red: 1, the arms: the two toothed arms in the upper half. 2, the zip stations: the three small holes along each arm.

Default	0.7 mm
Range	0 to 4 mm, in steps of 0.1 mm
Moves	arms, cord channel, finger lobes, zip stations

Arm tip width

Customizer name: **plate_arm_tip_width**

Widens the rounded tip of each arm, so the arms taper less.



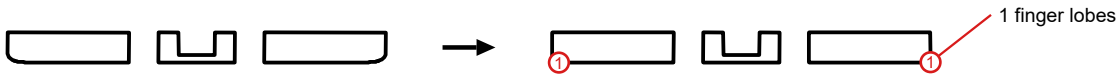
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: arm tip width at 16 mm. Marked in red: 1, the arms: the two toothed arms in the upper half. 2, the zip stations: the three small holes along each arm.

Default	11 mm
Range	5 to 16 mm, in steps of 0.5 mm
Moves	arms, zip stations

Outer edge rounding

Customizer name: **plate_edge_rounding**

Rounds or squares the outer face's edge all around the plate, while the plug-contact face stays square.



1.2 → 0 mm
section at y = 3 mm

Two vertical slices of the two-sided puller at y = 3 mm, before left and after right, the top face up. Left: the defaults, the plug in teal in the cut. Right: outer edge rounding at 0 mm. Marked in red: 1, the finger lobes: the two rounded lobes in the lower half. The plate edge, the arms, the teeth and the cord channel stay where they were.

Default	1.2 mm
Range	0 to 2 mm, in steps of 0.1 mm
Moves	finger lobes

Cable strip thickness

Customizer name: **plate_strip_thickness**

Thickens or removes the thin strip that bridges the cord channel on the outer face and ties the two arms together.



1 → 0 mm
section at y = 3 mm

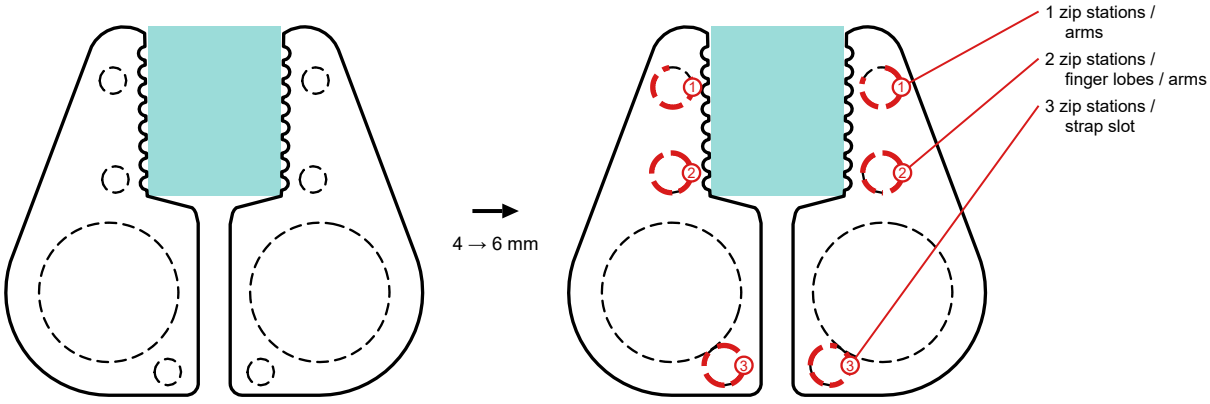
Two vertical slices of the two-sided puller at y = 3 mm, before left and after right, the top face up. Left: the defaults, the plug in teal in the cut. Right: cable strip thickness at 0 mm. Marked in red: 1, the cord channel: the gap between the lobes at the bottom. The plate edge, the arms, the teeth and the finger lobes stay where they were.

Default	1 mm
Range	0 to 4 mm, in steps of 0.25 mm
Moves	cord channel

Zip-tie hole diameter

Customizer name: **plate_zip_hole_diameter**

Widens the three zip-tie holes in each arm, and the stations shift to keep their walls.



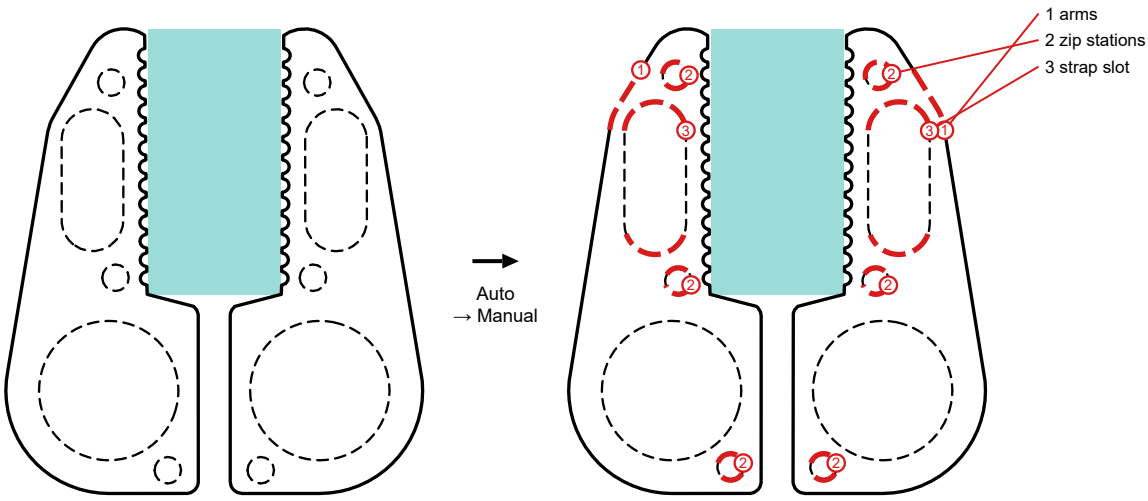
Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: zip-tie hole diameter at 6 mm. Marked in red: 1, the zip stations and the arms. 2, the zip stations, the finger lobes and the arms. 3, the zip stations and the strap slot.

Default	4 mm
Range	0 to 8 mm, in steps of 0.1 mm
Moves	arms, finger lobes, strap slot, zip stations

Zip-tie station placement

Customizer name: **plate_zip_placement**

Switches the three zip-tie stations in each arm from the automatic positions to the three position dials.



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40, a 20 mm wide plug in teal. Right: zip-tie station placement set to Manual. Marked in red: 1, the arms: the two toothed arms in the upper half. 2, the zip stations: the three small holes along each arm. 3, the strap slot: the long slot in each arm.

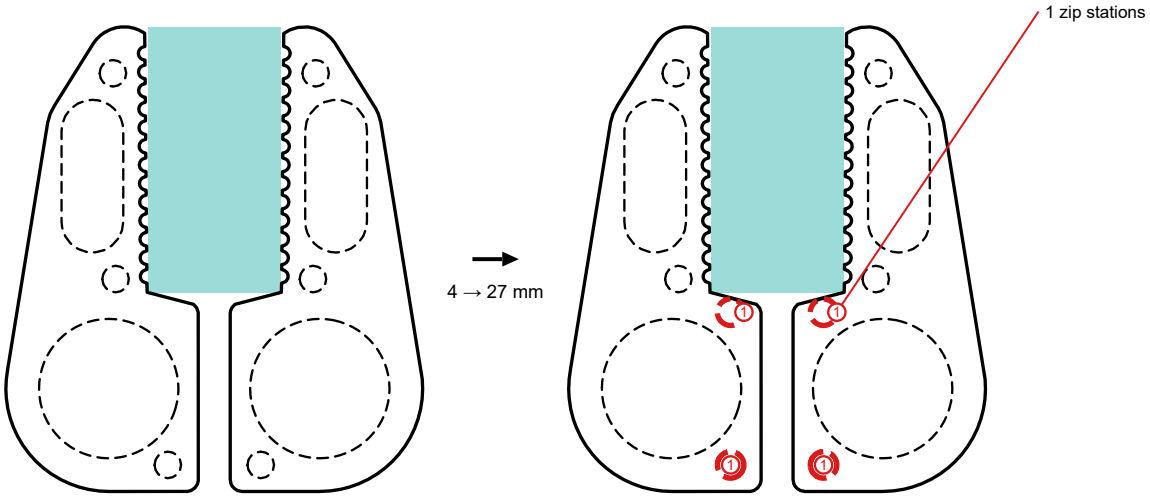
Default	Auto
Choices	Auto, Manual
Moves	arms, strap slot, zip stations

> **Note:** Drawn on a 40 mm plug so the default manual positions sit on the arm.

Zip-tie station 1 position

Customizer name: **plate_zip_pos_1**

Slides the rear zip-tie station along each arm, measured from the cord end.



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40 and zip-tie station placement set to Manual, a 20 mm wide plug in teal. Right: zip-tie station 1 position at 27 mm. Marked in red: 1, the zip stations: the three small holes along each arm.

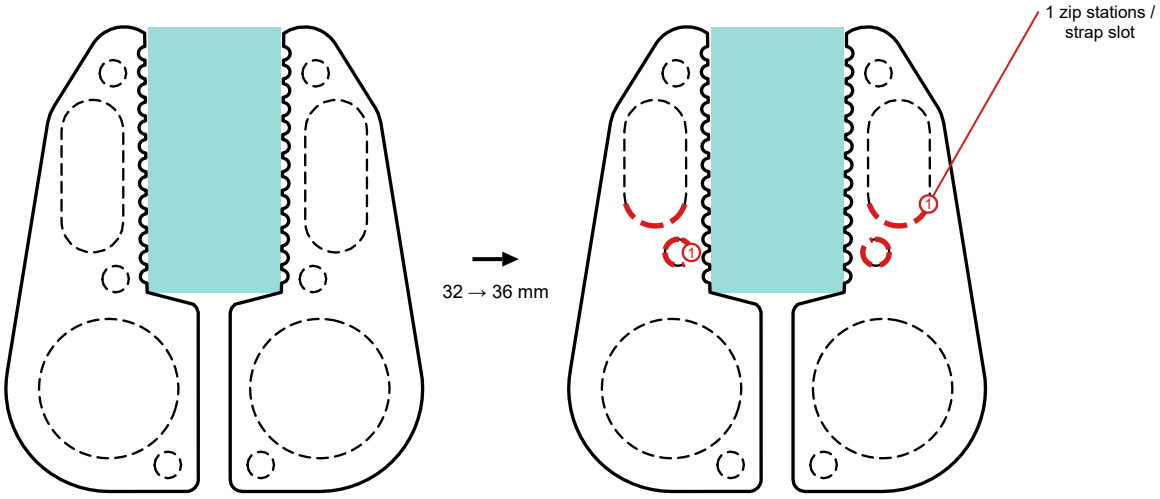
Default	4 mm
Range	0 to 80 mm, in steps of 0.5 mm
Moves	zip stations
Red warnings	ZIP STATION OFF THE ARM · ZIP STATIONS OVERLAP EACH OTHER · STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP

Note: Only acts when plate_zip_placement is Manual; drawn on a 40 mm plug so every station sits on the arm.

Zip-tie station 2 position

Customizer name: **plate_zip_pos_2**

Slides the middle zip-tie station along each arm, and the strap slot's window moves with it.



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40 and zip-tie station placement set to Manual, a 20 mm wide plug in teal. Right: zip-tie station 2 position at 36 mm. Marked in red: 1, the zip stations and the strap slot: the three small holes along each arm; the long slot in each arm.

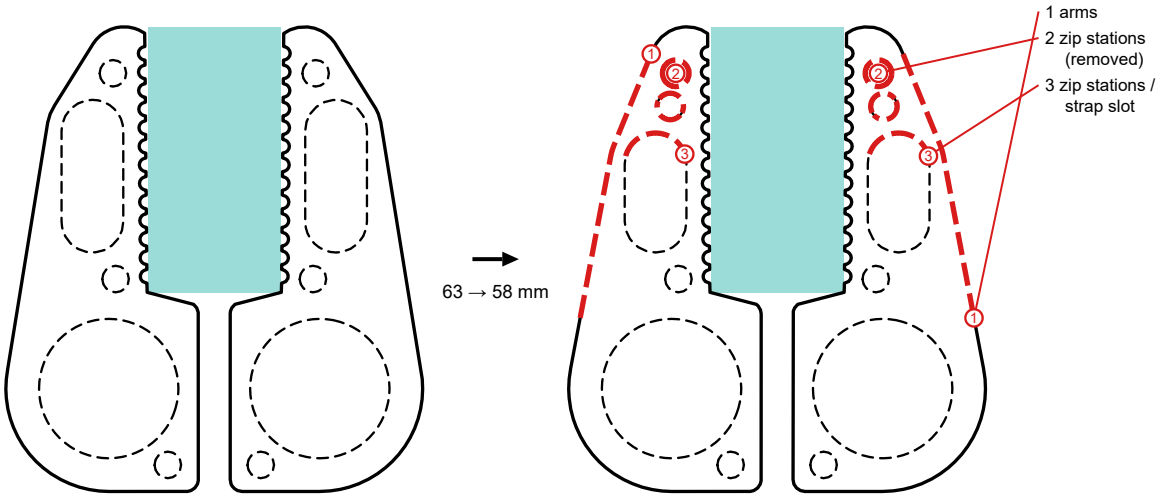
Default	32 mm
Range	0 to 80 mm, in steps of 0.5 mm
Moves	strap slot, zip stations
Red warnings	ZIP STATION OFF THE ARM · ZIP STATIONS OVERLAP EACH OTHER · ZIP STATION HITS VELCRO SLOT · STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP

Note: Only acts when plate_zip_placement is Manual; drawn on a 40 mm plug so every station sits on the arm.

Zip-tie station 3 position

Customizer name: **plate_zip_pos_3**

Slides the tip zip-tie station along each arm, and the strap slot's window moves with it.



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40 and zip-tie station placement set to Manual, a 20 mm wide plug in teal. Right: zip-tie station 3 position at 58 mm. Marked in red: 1, the arms: the two toothed arms in the upper half. 2, the zip stations: the three small holes along each arm, removed. 3, the zip stations and the strap slot.

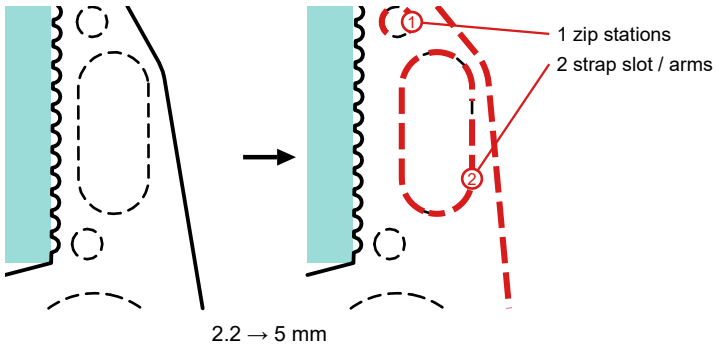
Default	63 mm
Range	0 to 80 mm, in steps of 0.5 mm
Moves	arms, strap slot, zip stations
Red warnings	ZIP STATION OFF THE ARM · ZIP STATIONS OVERLAP EACH OTHER · ZIP STATION HITS VELCRO SLOT · STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP

Note: Only acts when plate_zip_placement is Manual; drawn on a 40 mm plug so every station sits on the arm.

Wall between the teeth and the slot

Customizer name: **plate_slot_inner_wall**

Thickens the wall between the toothed edge and the strap slot, moving the slot outward.



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40, a 20 mm wide plug in teal. Right: wall between the teeth and the slot at 5 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the strap slot and the arms: the long slot in each arm; the two toothed arms in the upper half.

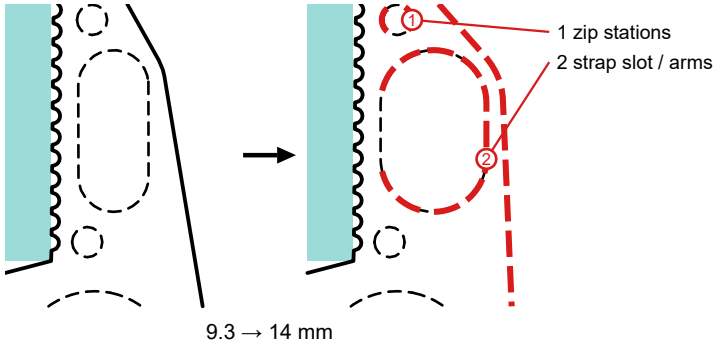
Default	2.2 mm
Range	1 to 8 mm, in steps of 0.1 mm
Moves	arms, strap slot, zip stations

> **Note:** Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot.

Strap slot width

Customizer name: **plate_velcro_slot_width**

Widens the strap slot in each arm, and the arm bulges outward to keep its wall.



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40, a 20 mm wide plug in teal. Right: strap slot width at 14 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the strap slot and the arms: the long slot in each arm; the two toothed arms in the upper half.

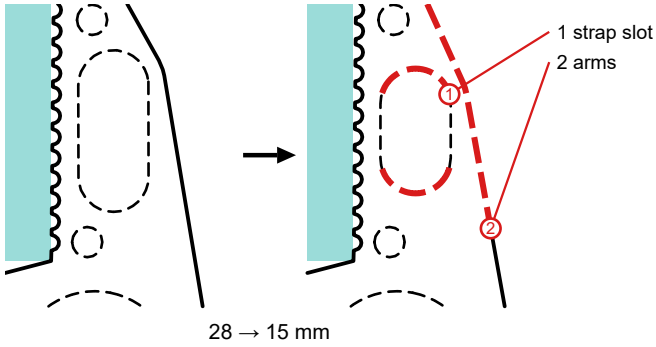
Default	9.3 mm
Range	0 to 20 mm, in steps of 0.25 mm
Moves	arms, strap slot, zip stations

> **Note:** Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot.

Strap slot length

Customizer name: **plate_velcro_slot_length**

Lengthens or shortens the strap slot along each arm, within the window between the zip-tie stations.



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40, a 20 mm wide plug in teal. Right: strap slot length at 15 mm. Marked in red: 1, the strap slot: the long slot in each arm. 2, the arms: the two toothed arms in the upper half.

Default	28 mm
Range	5 to 60 mm, in steps of 1 mm
Moves	arms, strap slot, zip stations
Red warnings	STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP

> **Note:** Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot.

Advanced - Render Quality

Render quality

Customizer name: **quality**

Sets how many flat segments draw each curve and changes no dimension.

> **No picture.** Changes no shape: it only sets the number of segments in every circle and arc.

Default	64 segments
Range	24 to 128 segments, in steps of 8 segments

Render, export and print

1. On the computer, press **F6** (or **Design ▸ Render**) and wait for the progress bar to finish; a render takes from a few seconds to a minute. In Forge, the preview draws on its own: wait for **Preview ready** under the picture.
2. Look for red text beside the part. If there is any, read it: it names the measurement to fix, and the part will not fit until it is gone.
3. Save the file: on the computer, **File ▸ Export ▸ Export as STL**; in Forge, **Export STL**. If you export without rendering, OpenSCAD asks you to render first: press F6 and export again. The quick preview (F5) looks complete but cannot be exported.

Load the STL into your slicer with these settings:

Setting	Value
Orientation	both plates flat on the bed, exactly as the file lays them out. No supports.
Layer height	0.2 mm
Walls	3 to 4: you pull hard on this part
Infill	25 to 35 percent
Material	PETG. PLA, ABS and ASA work too. Never a flexible filament: the tool has to stay stiff.

The file holds both plates side by side. Most slicers load it as one object with two parts, which prints fine as it is; use your slicer's split-to-objects command if you want to move one plate on its own.

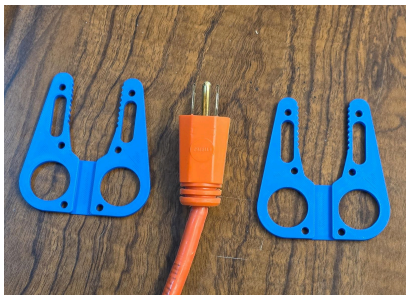
What to buy

Nothing to solder and almost nothing to buy: filament, zip ties, and a strap if you want one. For the two-sided puller you need three zip ties up to 3.6 mm wide (about 200 mm long). The filament each tool uses, and the strap widths that fit, are in the bill of materials.

Assemble

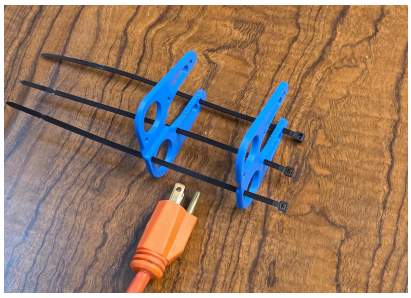
You need three zip ties; the size is in the bill of materials. The two plates are identical; one is flipped over so the toothed arms face each other.

1. Print both plates. The picture shows the two plates on a table with the plug between them.



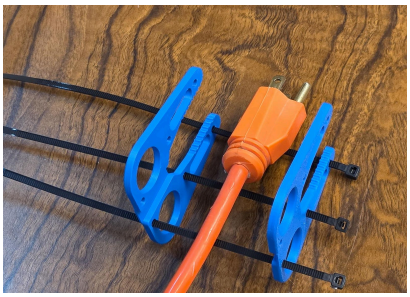
The two plates of the two-sided puller on a table with the orange extension-cord plug between them, ready to assemble.

2. Thread one end of each zip tie through the first plate, then through the second, so the ties lie between the plates. Make sure the cord channel of both plates faces the same way. The picture shows three ties through one plate, lying toward the other.



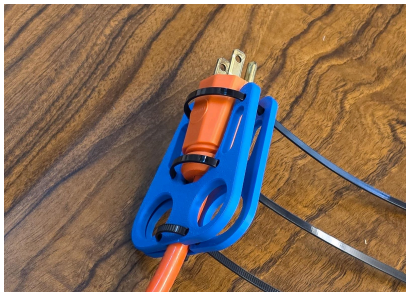
Three zip ties threaded through one plate of the two-sided puller and lying flat toward the second plate, the plug beside them.

3. Place your plug between the two plates, its prongs toward the arms. The picture shows the plug lying on the first plate between the ties.



The orange plug placed on the first plate between the zip ties, its prongs toward the plate's arms.

4. Thread the zip ties through the matching holes of the second plate. The picture shows the second plate over the plug with the ties through it and their tails loose.



The second plate laid over the plug, with the zip ties threaded through its matching holes and their tails hanging loose.

5. Pull the zip ties tight enough to hold the plug between the plates. The picture shows the ties cinched with their tails still on.



The zip ties pulled tight around both plates and the orange plug, their tails still sticking out.

6. Cut off the zip-tie tails. The picture shows the finished tool holding the plug.



The finished two-sided puller with the zip-tie tails cut off, holding the orange plug with its prongs clear.

The tool stays on the plug: the serrated arms do the gripping, and the zip ties hold the two plates together.

Use it

The plug puller is for anyone who cannot grip a plug and pull it out of an outlet: arthritis, low grip strength, tremor, a small hand, one hand. Two finger holes take the pull, so your whole hand does the work instead of a fingertip pinch. The tool touches only the plug's sides and back, never the outlet.

The two plates stay zip-tied around the plug, so there is nothing to seat: the tool is already on.

1. Put two fingers through the round holes at the cord end, one in each lobe.
2. Pull straight back, away from the wall.

The pictures show a two-sided puller and its plug seated in an outlet, ready to be pulled, and then the pull.



The two-sided puller and its orange extension-cord plug seated in a wall outlet, ready to be pulled, no hand on it yet.



A gloved hand pulling the two-sided puller and its orange extension-cord plug straight out of a wall outlet, two fingers through the holes.

A two-sided puller on a USB-C laptop tip works the same way, pulling the tip out of the laptop instead of a wall.

Safety

- Nothing goes between the plug face and the outlet cover: the tool grips only the plug's sides and back.
- Pull straight out. Never lever the tool sideways or up and down.
- Do not use the tool on a damaged cord, a cracked plug, or a plug that is warm to the touch.
- Keep your fingers away from the prongs as the plug comes out.
- The tool has no conductive parts, but it is still plastic near electricity: if it cracks, stop using it.

Care

- Before each use, look at the finger holes and the places the zip ties or the strap pass through for cracks. A cracked tool is replaced, not repaired.
- Replace a zip tie that has loosened.
- Wash with soap and warm water. No solvents: they attack PETG and PLA.
- Keep it out of direct sun and off heaters; the plastic softens when hot.

If the print does not fit

Printed your plates and something is not quite right? Every fix is one dial. Change it in the Customizer, render and export again, and reprint.

> Nudge in small steps, 0.5 to 2 mm. Looser always beats tighter: a slightly roomy tool still works, a tight one does not.

Symptom	Change this
The plug will not drop in between the arms, or the plates will not close	Plug width at the end that binds, +1 mm. On Rounded sides, a larger Grip clearance also loosens the fit
The plug slides out of the closed plates	Flat sides: a more negative Grip bite, −1 to −2 mm, so the arms squeeze. Rounded sides: a deeper Cradle depth
The cord is pinched in the channel, or the plates will not close around it	Cord thickness +0.5 mm, or a larger Cord channel clearance
Fingers pinch in the holes, or a knuckle drags on the rim	Hand size up one step, or Measure my hand with the finger width +1 mm; or a larger Finger hole fit
Fingers swim in the holes	Hand size down one step, or Measure my hand with the finger width −1 mm
A plate flexes under load, or feels flimsy	A larger Extra wall on the plate, or a thicker Plate thickness
The arms are shorter than the plug, or much longer	Plug length: measure from the surface it plugs into to the plug's back end, not including the cord

The dials named here are in **Advanced - Two-Sided Puller**, described one by one in this guide. In version 0.14.0 the four presets held their plugs loosely in test prints: their numbers had come from other sample plugs, and the gap narrowed in a straight line from the prong end, so a plug with a wide head touched the teeth only at the head's back corners. From version 0.15.0 the presets carry numbers measured on their own plugs and a head length that keeps the gap at the prong-end width along the head. Reports from plates printed on the new presets go here when they come in.

Red warning tags

If the preview shows red text, or a red text tag printed next to your part, the file found a problem before you wasted a full print. That is on purpose: a bad file fails loudly instead of silently. Read the tag, fix that measurement, and export again.

The tag says	What it means	What to do
`CHECK PLUG LENGTH MEASUREMENT (MM?)`, and the same for `PLUG WIDTH`, `CORD THICKNESS`, `FINGER WIDTH`	That number is outside any plausible mm value: usually inches typed into a mm field (1.25 instead of 32)	Measure again with the mm side of the ruler and type it again
`CORD TOO THICK FOR CABLE CHANNEL`	The cord will not fit the channel with clearance	Measure the cord thickness again, or raise the cord channel clearance
`PLUG TOO WIDE - ARMS BULGE PAST FINGER LOBES`	The plug is so wide that the arms would bulge wider than the finger lobes	Check the plug width; a truly huge plug is beyond this tool
`NO GRIP BITE - PLUG WONT BE HELD`	Flat sides or a plug preset with a grip bite of 0 or more, so the arms do not squeeze	Set the grip bite negative, for example −1, so the arms bite the plug
`PLATE THINNER THAN 2MM - TOO FLIMSY`	The plate thickness is under 2 mm	Raise the plate thickness to 3 to 5 mm
`ZIP STATION OFF THE ARM`	A manual zip-tie station fell past the arm	Bring that station position back within the arm's length
`ZIP STATIONS OVERLAP EACH OTHER`	Two manual zip-tie stations landed on top of each other	Spread the station positions at least one hole diameter apart
`ZIP STATION HITS VELCRO SLOT`	Manual placement only: a station broke into the strap slot	Move station 2 or 3; in Auto placement the slot always keeps clear of the stations
`PLUG TOO LONG - PLATE OVER 120MM, CHECK PLUG LENGTH`	The arms grow with the plug, and this plug length pushed the plate past 120 mm	Measure the plug length again, from the surface it plugs into to the plug's back end, not including the cord
`PLUG WIDTH TAPER LOOKS WRONG - RECHECK BOTH ENDS`	The two widths are more than about 20 degrees of taper apart, almost certainly a mistake	Measure the plug width at the prong end and at the cord end again
`STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP`	Manual placement only: the strap width is wider than the slot the arm can offer between the stations	Use a narrower strap, or raise the strap slot length or move the stations to widen the window
`PLUG NARROWER THAN THE CORD CHANNEL - ARMS CANNOT TOUCH IT`	At one end the plug is narrower than the cord channel, so the arms cannot reach it there	Measure the plug widths and the cord itself again, not its strain relief; a smaller cord channel clearance narrows the channel
`CRADLE SHALLOWER THAN ASKED - PLUG NARROW`	Rounded sides: the plug is so narrow that the full cradle would close below the cord channel, so it was built shallower	A smaller cradle depth, or measure the cord itself, not its strain relief
Orange `STRAP SLOT LEFT OUT - PLUG TOO SHORT FOR ONE` (preview only)	Auto placement: the plug is too short for a strap slot, so the arms have none; the zip ties still hold	Nothing to fix; pick Zip ties in Step 3 and the note goes away

Printing problems

Symptom	Cause	Fix
The slicer complains about floating or disconnected parts	You exported with a red warning tag showing; the tag is a separate object	Fix the named measurement first
The slicer shows two separate parts	That is the design: Both plates puts the two plates side by side in one file	Print both; if the slicer asks, let it split them into separate objects
Layers split at the pocket floor or around a hole	Under-extrusion, or too few walls	3 to 4 walls and 25 percent infill or more
The tool snapped while pulling	A brittle material, or walls too thin	Reprint in PETG with 4 walls, or raise plate_wall_boost or plate_thickness

Advanced settings

Everything below Step 4 in the Customizer is optional. The **Advanced - Two-Sided Puller** section holds the plate dials, which work with your measured sizes and apply in every hand size; there is no Custom size in this file. Every one of these dials is described, with a picture, earlier in this guide.

How the plate is sized

The plate takes its main sizes from Step 1. The arm's gripping edge follows the plug's own two widths, the prong-end width at the arm tips and the cord-end width at the plug's back end, blended in between; with a plug head length, the edge holds the prong-end width along the head and blends only behind it. The arms grow so they always cover the whole plug length. The throat, where the gap closes down to the cord channel, sits just below the plug's back end. The cord channel comes from the cord thickness, and the finger holes from the Step 2 hand size. The plate dials tune that result; their defaults carry the field-tested grip: 2 mm teeth on a 2.8 mm pitch biting 1 mm deep, a grip bite of −1, with the toothed zone set automatically to the whole plug body.

Step 3's attachment offers Zip ties + Velcro strap (the default) and Zip ties. The three zip-tie stations per arm are always there, because the ties are what cinch the two plates together; the strap slot in each arm comes with the strap choice. The strap width sets the slot's length so the strap always threads through. In Auto placement, a plug too short for a slot gets none, with the orange preview note `STRAP SLOT LEFT OUT - PLUG TOO SHORT FOR ONE`.

Rounded or flat sides: the cradle

Step 1's plug sides choice picks how the arms meet the plug. **Rounded sides**, the default, builds a sloped cradle: the gap between the arms is narrower at each plate's outer face than where the plates meet, by the cradle depth per side (2.5 mm by default, so the gap narrows by 5 mm), and the grip clearance (0.5 mm) is the extra room where the plates meet, so a hard plug drops in flat and the cradle does the holding. Stacked face to face, the two plates form a diamond-shaped channel that centers a round plug, a USB-C tip or a charger tip. The teeth follow the slope.

The cradle never makes the outer-face gap narrower than the cord channel. When a narrow plug would need that, the file builds a shallower cradle and shows `CRADLE SHALLOWER THAN ASKED - PLUG NARROW`; a smaller cradle depth, or measuring the cord itself rather than its strain relief, clears it.

Flat sides keeps the straight toothed edge, and there the grip bite sets the squeeze. The cradle depth and the grip clearance apply to Rounded sides only, and the grip bite to Flat sides. The plug presets carry their own sides; the USB-C laptop tip's Rounded sides also carry their own cradle depth, 1.9 mm, the most that fits beside its cord channel.

Print layout

Step 4's print layout is Both plates by default: the two identical plates side by side in one file, 8 mm apart at their widest points, so one export prints the whole tool. One plate exports a single plate. Either way, flip one plate over after printing and zip-tie the pair face to face around the plug.

Strength

Goal	Dial
A denser, stronger plate all around	Extra wall on the plate: every wall around the inner openings grows by that much; the holes move with it
A stiffer sandwich	Plate thickness (each plate; the pair is twice that)
A thicker wall between the teeth and the strap slot only	Wall between the teeth and the slot, measured from the deepest tooth bite, so bigger teeth never silently thin it
Less plastic	A wider or longer strap slot (the slots double as material reduction), or a cable strip thickness of 0

Grip

Goal	Dial
A steeper or gentler cradle (Rounded sides)	Cradle depth, per side; 0 is a straight edge
More or less room where the plates meet (Rounded sides)	Grip clearance, in total, not per side
Squeeze harder or looser (Flat sides and the presets)	Grip bite; negative squeezes, applied per side on top of the plug's own widths
Teeth that bite a soft plug body	Tooth size, tooth spacing and tooth depth
Where the teeth sit	Toothed zone start and toothed zone length, measured back from the arm tips; a length of 0 covers the whole plug body
Easier plug entry	Tip flare, which opens the gap at the tips
A cord that slides freely	Cord channel clearance
Finger security	Finger hole fit: the hole is the finger width plus this

Manual zip-tie placement, with the three station position dials, trades the collision guarantees for full control; the red tags name a station that falls off the arm, overlaps another, or breaks into the strap slot.

The full list of dials with their ranges, steps and defaults is the machine-readable file `parameter_mapping_two_sided.json`, checked against the model by the project's tests.

Guard rails

The model never silently changes your input. Anything truly wrong prints a red warning tag flat on the bed beside the part, so an exported file shows its own defect, and writes the same message to the console. The see-through plug and the orange strap-slot note appear in the preview only and are never exported.

Saved settings

The Customizer panel has a preset bar above the sections: the **+** button saves your current values as a named set inside a JSON file next to the `.scad` file. The sets that ship with the project are in `presets/Plug_Puller_Two_Sided.json`. They are plain text: easy to keep, compare and share.

The command line

Every dial can be set from the command line, which makes batches and A/B tests scriptable:

```
# Both plates for the heavy-duty extension cord preset, with 2 mm thicker walls
openscad -o plates.stl --backend Manifold \
  -D 'plug_preset="Heavy-duty extension cord - NEMA 5-15"' -D plate_wall_boost=2 \
  src/Plug_Puller_Two_Sided.scad

# A saved set of settings
openscad -o plates.stl -p presets/Plug_Puller_Two_Sided.json \
  -P "USB-C laptop tip" src/Plug_Puller_Two_Sided.scad
```

Going deeper

- The one-sided puller has its own quick start and full guide, in the folder beside this one.
- The design rationale: why each default is what it is, with its number and its source.
- The engineering reference: both tools' geometry, coordinate frames, the derivation layer, render modes, the console messages and every validation check.
- The dial pictures: every picture in this guide with its text alternative, and how the pictures are described.
- The bill of materials: zip ties, the strap, filament per tool.

Plug Puller — Measuring form, two-sided puller

Measure in mm. Fill in the blanks top to bottom, then type them into the Customizer in the same order.

Print at 100 % — 50 mm

1. `plug_preset`
   Plug preset
   - (●) **Measure my plug**
   - ( ) Heavy-duty extension cord - NEMA 5-15
   - ( ) USB-C laptop tip
   - ( ) Flat 2-prong lamp plug - NEMA 1-15
   - ( ) Standard 3-prong plug - NEMA 5-15

2. `measure_plug_length`
   Plug length
   default 25.5 mm → _____ mm card R1

3. `measure_plug_width_prong_end`
   Plug width at the prong end
   default 20 mm → _____ mm card R1

4. `measure_plug_head_length`
   Plug head length
   default 0 mm → _____ mm card R1

5. `measure_plug_width_cord_end`
   Plug width at the cord end
   default 20 mm → _____ mm card R1

6. `measure_cord_thickness`
   Cord thickness
   default 4 mm → _____ mm card C1

7. `plug_sides`
   Plug sides
   - (●) **Rounded sides**
   - ( ) Flat sides

8. `show_plug_preview`
   Show the see-through plug
   - (●) **leave on**

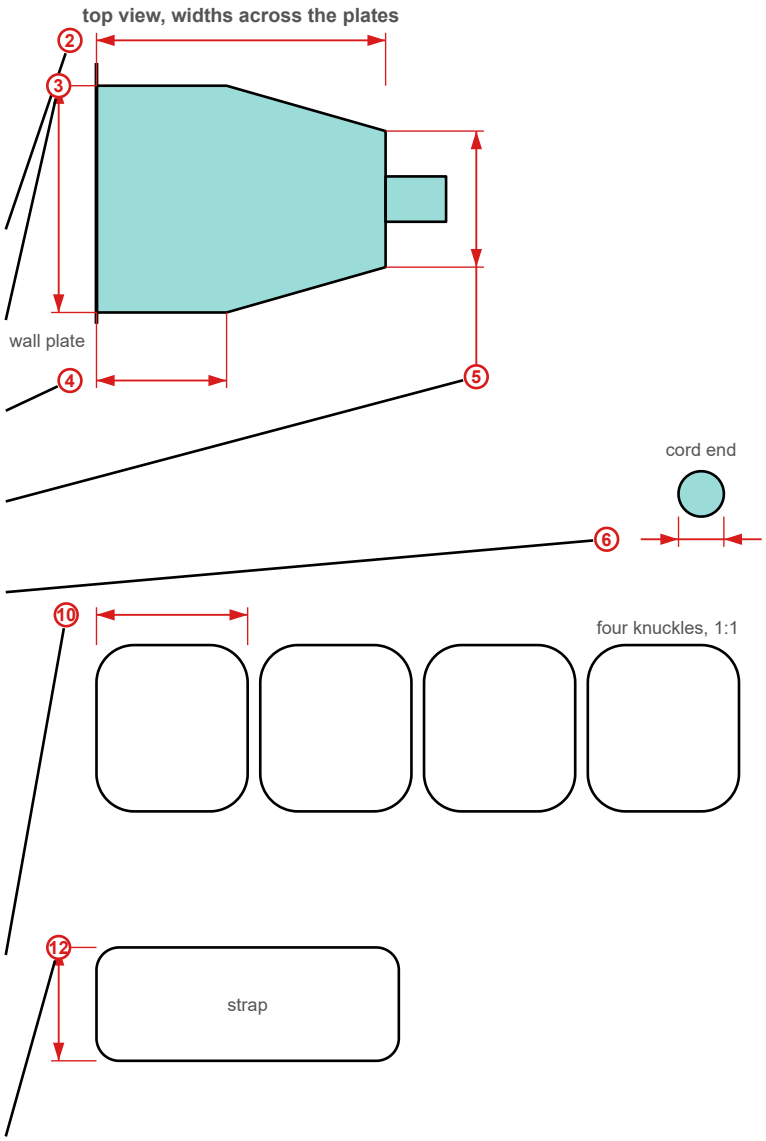
9. `size`
   Hand size
   - ( ) Small
   - (●) **Medium**
   - ( ) Large
   - ( ) Measure my hand

10. `measure_finger_width`
    Finger width
    default 20 mm → _____ mm card F1 / F2

11. `attachment`
    Attachment
    - (●) **Zip ties + Velcro strap**
    - ( ) Zip ties

12. `strap_width`
    Strap width
    default 15 mm → _____ mm

13. `print_layout`
    Print layout
    - (●) **Both plates**
    - ( ) One plate



Plug Puller — Measuring Stencil Sheet

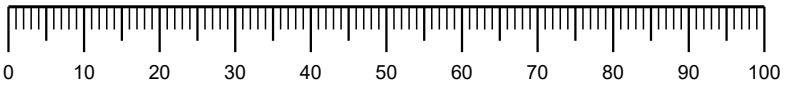
Print at 100% scale / “Actual size” — never “fit to page”. Check the calibration square before trusting anything on this sheet.

CALIBRATION
50 × 50 mm

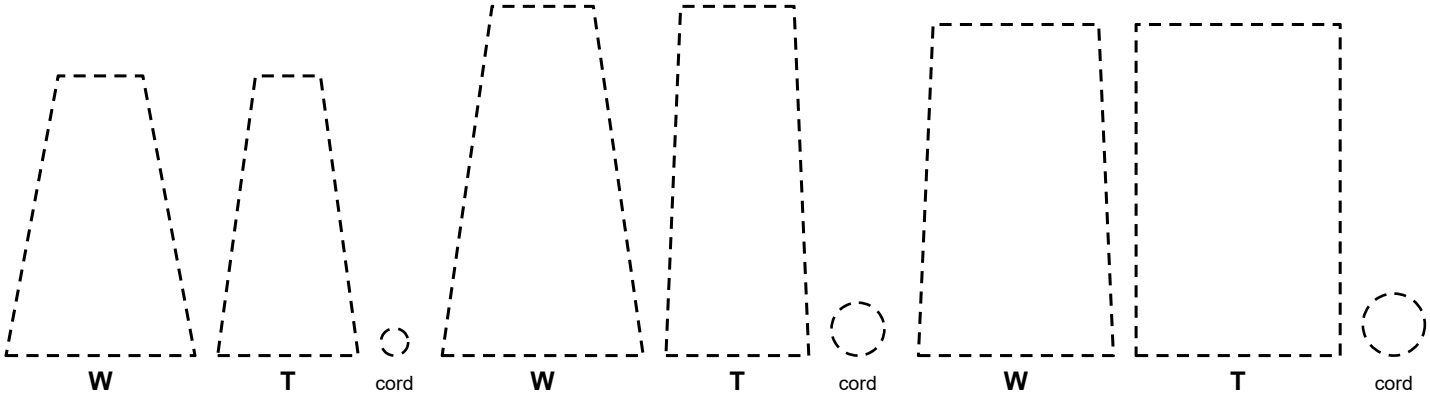
1. Measure this square with a ruler.
It must be exactly 50 × 50 mm — if not, your print was scaled: re-print at 100%.

Card legend: the quick start, under Match a card
Works on A4 and US Letter paper.

R1 — ruler (mm): plug length, widths, thicknesses, hand width



2. Plug silhouettes (1:1) — hold your plug against W (wide side) and T (thin side); if it matches, pick that preset in Step 1.



P1 Lamp 2-prong NEMA 1-15

P2 Standard 3-prong NEMA 5-15

P3 Heavy-duty cord NEMA 5-15

3. F1 / F2 — finger circles: smallest circle your middle finger passes through, minus 5 = finger width.

